Cross-Domain Generalization for Wearable-Based Human Activity Recognition

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Abstract

Human activity recognition (HAR) models trained on smartphone inertial data typically perform well on the population and device configuration they were trained on, but accuracy often degrades when they are applied to new users, a different sensor placement, or a different device. This is one instance of the broader *lab-to-field* generalization problem in mobile health. This proposal describes a project that uses two public smartphone HAR datasets, UCI HAR and WISDM, which differ in device, placement, sampling rate, and participants, to (1) quantify the generalization gap between and within datasets and (2) test whether lightweight, low-label adaptation strategies can close it.

1 Introduction

Activity recognition underpins many consumer and clinical mobile-health applications, from fall detection and step counting to sedentary-behaviour monitoring. In practice, however, a model is rarely deployed on the exact population, device, and sensor placement it was trained on. Understanding *how much* accuracy is lost under such domain shift, and *why*, determines whether HAR models can be trusted in real deployments rather than only on curated benchmarks.

Prior work in mobile sensing has shown that models trained in controlled, lab-like settings often rely on spurious cues (for example device orientation or individual gait idiosyncrasies) that do not transfer to new users or field conditions. Most HAR papers, however, report results on a single benchmark with a fixed split, and cross-dataset evaluation is comparatively less common.

Novelty. This project does not propose a new architecture. Its contribution is a controlled, diagnostic study: we (i) separate *subject shift* (unseen users within one dataset) from *cross-dataset shift* (different device, placement, sampling rate, and protocol) using the same models and preprocessing, (ii) identify which activities are most affected, and (iii) measure how many labelled target-domain samples a lightweight adaptation method needs to recover performance (a label-efficiency analysis). We present this as a specific gap we fill, not as a claim of a new state of the art.

2 Research Questions

We address the following research questions:

- **RQ1:** How much does HAR performance (accuracy and macro F1) degrade when a model trained on one dataset is evaluated on the other, compared with (a) in-domain evaluation and (b) evaluation on unseen subjects within the same dataset?
- **RQ2:** Which activities are most affected by domain shift, and which sources of shift (sensor placement, sampling rate, sensor set, or population) plausibly explain the errors?

- **RQ3:** Can lightweight, low-supervision adaptation (per-domain feature normalization or alignment, and fine-tuning on a small labelled target sample) recover the lost performance, and how many labelled target samples are needed?

3 Dataset Description

We will use two publicly available, pre-labelled smartphone HAR datasets, each serving as source and target in turn (UCI HAR $\rightarrow$ WISDM and WISDM $\rightarrow$ UCI HAR). No new data collection is required.

3.1 Source and collection

UCI HAR (Human Activity Recognition Using Smartphones) contains recordings from 30 volunteers aged 19–48 years. Each wore a Samsung Galaxy S II smartphone on the waist while performing six activities: walking, walking upstairs, walking downstairs, sitting, standing, and laying. The phone's accelerometer and gyroscope captured 3-axial linear acceleration and angular velocity at a constant 50 Hz. The sessions were video-recorded so that the data could be labelled manually. The signals were noise-filtered and cut into fixed-width windows of 2.56 s with 50% overlap (128 readings per window), and the release is partitioned by volunteer into 70% training and 30% test subjects.

WISDM (Wireless Sensor Data Mining Activity Prediction dataset, collected at Fordham University) contains tri-axial accelerometer data from a smartphone carried in the participant's front pants pocket, sampled at 20 Hz, for six activities: walking, jogging, walking upstairs, walking downstairs, sitting, and standing, comprising roughly 1.1 million samples.

	UCI HAR	**WISDM**
Placement	Waist	Front trouser pocket
Sensors	Accelerometer + gyroscope	Accelerometer only
Sampling rate	50 Hz	20 Hz
Activities	Walk, upstairs, downstairs, sit, stand, lay	Walk, jog, upstairs, downstairs, sit, stand

3.2 Suitability

The two datasets differ in phone model, placement, sampling rate, available sensors, activity set, and participants, so evaluating across them produces a genuine domain shift that is representative of moving a model from one deployment to another. Both are widely used, openly available, and use the same kind of sensor (a smartphone accelerometer), which makes harmonization feasible. To make them comparable we will: (i) use only the shared channel (tri-axial accelerometer) and discard the gyroscope; (ii) resample UCI HAR from 50 Hz to 20 Hz; (iii) use a common window length (for example 5 s with 50% overlap; configurable); and (iv) restrict to the five activities the datasets share (walking, upstairs, downstairs, sitting, standing), dropping jogging and laying. Data within a dataset will be split by subject.

3.3 Potential shortcomings

- **Confounded shift.** Placement, sampling rate, sensors, device, and population all change at once between these datasets, so we cannot cleanly attribute a performance drop to one factor. We will partly address this with controlled synthetic perturbations (for example rotation, noise, and down-sampling) applied within one dataset. Sitting vs. standing is especially fragile, since it depends on device orientation, which differs between waist and pocket.
- **Small, scripted cohorts.** Both datasets have only tens of participants performing scripted, lab-style activities, so results may not reflect free-living behaviour, and harmonizing to five classes discards jogging and laying.
- **Class imbalance.** WISDM is dominated by walking and jogging, with sitting and standing comparatively rare, so accuracy alone can mislead; we will report macro F1 and per-class results.
- **Leakage.** Windows from the same subject are highly correlated and overlapping windows share samples, so all splits must be by subject.

4 Proposed Approach

1. **Baseline models.** Train classical models (random forest or gradient-boosted trees over statistical and frequency-domain features) and a deep model (a 1D CNN or LSTM over raw windows) on the source dataset, using subject-wise splits. The final model family will be chosen based on early results.
2. **Shift evaluation.** Evaluate each model (a) on held-out subjects of the source dataset and (b) on the other dataset in full, to separate subject shift from cross-dataset shift.
3. **Adaptation strategies.** Test per-domain feature normalization or alignment (for example standardization or CORAL-style alignment) and fine-tuning on a small labelled target sample (k windows per class), varying k to obtain a label-efficiency curve.
4. **Error analysis.** Analyse per-activity errors, confusion patterns, and the effect of synthetic perturbations, to explain the quantitative results.

5 Evaluation Plan

We report accuracy, macro F1, per-class F1, and confusion matrices for in-domain, unseen-subject, and cross-dataset settings, in both transfer directions. We report the relative improvement from adaptation over the unadapted cross-dataset baseline, and the label-efficiency curve as a function of k. Variability is assessed with repeated subject-wise splits (or subject-level cross-validation) and reported as mean $\pm$ standard deviation.

6 Execution Plan

Timeframe	Milestone
Weeks 1–2	Acquire and preprocess both datasets; harmonize channels, sampling rate, windows, and labels; implement feature extraction.
Weeks 3–4	Train and tune classical and deep baselines; report in-domain and unseen-subject performance.
Weeks 5–6	Run cross-dataset evaluation in both directions; quantify the generalization gap (progress report checkpoint).
Weeks 7–9	Implement and evaluate adaptation strategies and the label-efficiency analysis.
Weeks 10–11	Consolidate results, error analysis, and ablations with synthetic perturbations.
Week 12	Prepare final presentation and final report.

7 Expected Outcomes and Risks

The expected deliverable is a quantitative characterization of HAR generalization gaps, separating subject shift from cross-dataset shift, together with evidence on whether lightweight adaptation can close them with few labels. The main risks are the confounded shift between these two datasets (mitigated with synthetic perturbations) and the reduced task after harmonization (accelerometer only, five classes), which we will state when interpreting results.

8 Team Justification

We request approval of a three-person team because the work splits into three substantial, largely independent workstreams: harmonizing and validating two datasets, evaluating two model families in two transfer directions, and an adaptation study with a label-efficiency sweep. **Andrew Goetz** owns data and preprocessing (harmonization, subject-wise splits, leakage checks). **Shubham Sharma** owns models and shift evaluation (classical and deep baselines; in-domain, unseen-subject, and cross-dataset evaluation in both directions). **Siddharth S. Taragi** owns adaptation and analysis (adaptation methods, label-efficiency sweep, synthetic perturbations, per-activity error analysis). All three share the report and presentation. Because we are in the same program, we share coursework, tools, and schedules, which makes coordination, code review, and frequent check-ins easier and lowers integration risk.